SIMATS ENGINEERING
ASSESSMENT TOOL

COURSE NAME: Theory of Computation **COURSE CODE: CSA1304**

COURSE OUTCOME COVERED

CO1: Analyze fundamental mathematical concepts of automata theory for formal languages and decision problems.(BL:3)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:25**
Duration : 1 Hour

Analytical Problem

Code of Conduct:

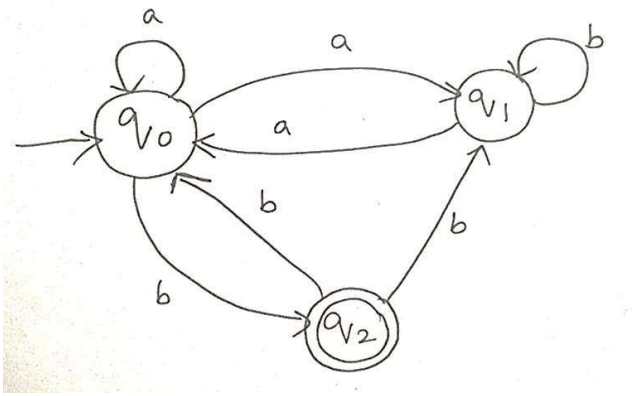
I GIFTSON KEVIN MATTHEW.Y (192511167)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: GIFTSON KEVIN MATTHEW.Y

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1. Consider the NFA given below which accepts some language L over the alphabet **Σ**={a,b}. Construct a DFA that represents the same language L

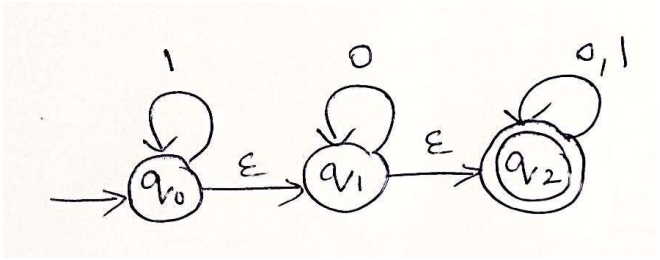


2. Let L be the language of words of length atleast 2 whose last two letters are difference from each other. For example abaab is in L but not bb, Construct an NFA for L, Then draw the equivalent DFA

3. Design a Nondeterministic Finite Automata (NFA) that takes a binary string as input and accepts the string only if the last two digits are the same (either 00 or 11). Write the formal definition of the NFA. Also show that the string **01011** is accepted by the NFA

4. Consider the NFA given below which accepts the language representing the set of all strings having 001 as a substring over the alphabet **Σ**={0,1}. Construct a DFA that represents the same language.

5. A NFA with ε-transitions is given below. Find ε-closure for each state and Design an equivalent DFA which accepts the same language. Draw its transition table and mark the initial and final states



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RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**10**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

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Question 1. Convert the given NFA into an equivalent DFA

Given: Alphabet $\Sigma = \{a, b\}$.

From the diagram, the NFA has three states: q_0, q_1, q_2. The initial state is q_0, and the final state is q_2.

Step 1: Write the NFA transition table

State	a	b
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_2\}$
q_1	$\{q_0\}$	$\{q_1\}$
$* q_2$	$\emptyset$	$\{q_0, q_1\}$

Step 2: Apply subset construction

Let the DFA states be sets of NFA states.

- $A = \{q_0\}$— initial state
- $B = \{q_0, q_1\}$
- $C = \{q_2\}$
- $D = \{q_1, q_2\}$
- $E = \emptyset$— dead state

Calculate the transitions:

1. From $A = \{q_0\}$:

- On a: $\{q_0, q_1\} = B$
- On b: $\{q_2\} = C$

2. From $B = \{q_0, q_1\}$:

- On a: $\{q_0, q_1\} = B$
- On b: $\{q_1, q_2\} = D$

3. From $C = \{q_2\}$:

- On a: $\emptyset = E$
- On b: $\{q_0, q_1\} = B$

4. From $D = \{q_1, q_2\}$:

- On a: $\{q_0\} = A$
- On b: $\{q_0, q_1\} = B$

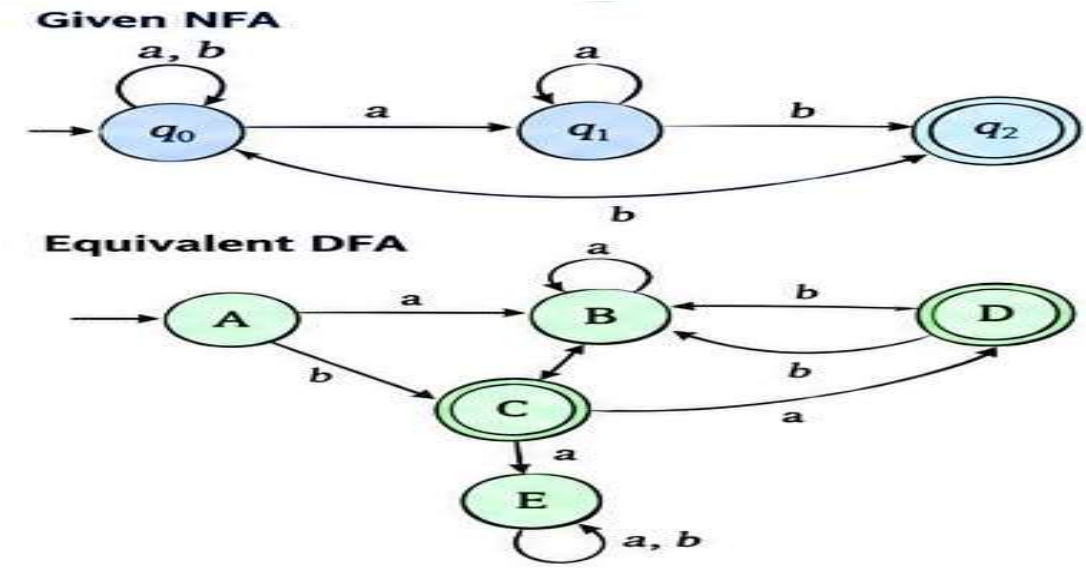
5. From $E = \emptyset$:

- On both $a$and b: E

Step 3: Equivalent DFA transition table

DFA state	a	b
$\rightarrow A$	B	C
B	B	D
$* C$	E	B
$* D$	A	B
E	E	E

The final states are $C$and D, because they contain the NFA final state q_2.



Result: The equivalent DFA is obtained using subset construction, and it accepts the same language as the given NFA.

Question 2. Construct an NFA and equivalent DFA for strings of length at least 2 whose last two letters are different

Given: $\Sigma = \{a, b\}$.

The language contains strings ending in either ab or ba. Examples: ab, ba, aab, bba, abaab.

Strings such as aa, bb, and a are not accepted.

A. NFA construction

Define the NFA as

$$N = (Q, \Sigma, \delta, q_0, F)$$

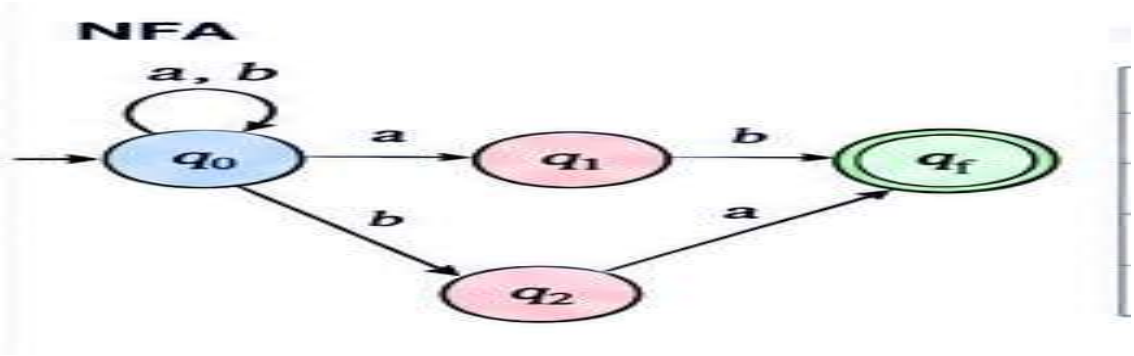
where:

- $Q = \{q_0, q_1, q_2, q_f\}$
- $\Sigma = \{a, b\}$
- Initial state: q_0
- Final state: $F = \{q_f\}$

Transition table:

State	a	b
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0, q_2\}$
q_1	$\emptyset$	$\{q_f\}$
q_2	$\{q_f\}$	$\emptyset$
$* q_f$	$\emptyset$	$\emptyset$

Explanation: State $q_0$reads any prefix. The NFA can guess that the current letter is the second-last letter. It accepts when the next letter is different.



For clarity in your exam, draw a self-loop on $q_0$labelled a, b, an arrow $q_0 \xrightarrow{a} q_1$, an arrow $q_0 \xrightarrow{b} q_2$, and the two accepting paths shown above.

B. Equivalent DFA

The DFA remembers the most recent letter and whether the string has ended with two different letters.

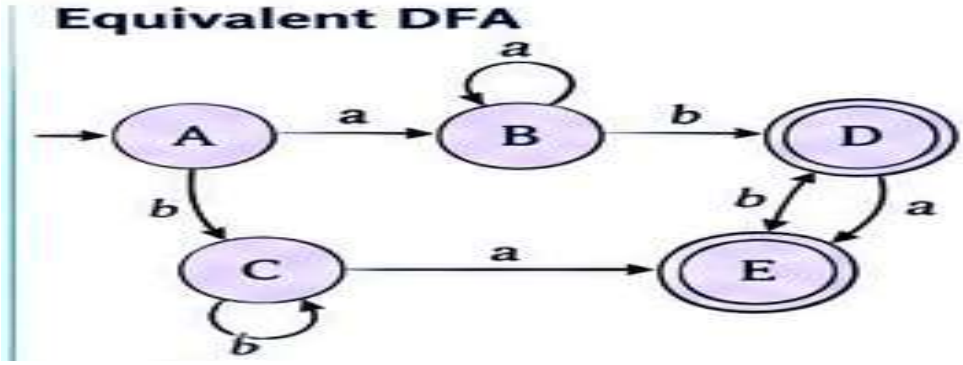
Define the states:

- A: no letter read yet
- B: last letter is a, not accepted yet
- C: last letter is b, not accepted yet
- D: last two letters are different and end in ab
- E: last two letters are different and end in ba

Transitions are determined by the latest two letters.

DFA state	a	b
$\rightarrow A$	B	C
B	B	D
C	E	C
$* D$	E	C
$* E$	B	D

Final states: D, E.



Result: The NFA and DFA accept exactly the strings whose last two letters are different.

Question 3. Design an NFA that accepts binary strings whose last two digits are the same

Given: The alphabet is $\Sigma = \{0, 1\}$. The string must end in either 00 or 11.

Examples:

- Accepted: 00, 11, 100, 011, 01011
- Rejected: 01, 10, 101, 110

A. Formal definition of NFA

The NFA is defined as

$$N = (Q, \Sigma, \delta, q_0, F)$$

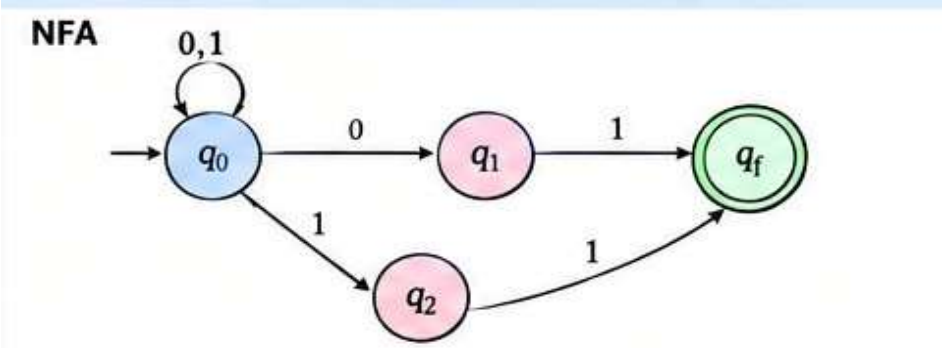
where:

- $Q = \{q_0, q_1, q_2, q_f\}$
- $\Sigma = \{0, 1\}$
- q_0 is the initial state

- $F = \{q_f\}$is the set of final states

The transition function $\delta$is given below.

State	0	1
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0, q_2\}$
q_1	$\{q_f\}$	$\emptyset$
q_2	$\emptyset$	$\{q_f\}$
$* q_f$	$\emptyset$	$\emptyset$



Draw the self-loop on $q_0$with label 0,1. The NFA guesses where the last two digits begin.

C. Show that the string 01011 is accepted

Input string:

$$w = 01011$$

Follow one valid path:

$$q_0 \xrightarrow{0} q_0 \xrightarrow{1} q_0 \xrightarrow{0} q_0 \xrightarrow{1} q_2 \xrightarrow{1} q_f$$

The last two digits are 11, so the NFA reaches the final state q_f.

Result: The string 01011 is accepted because its last two digits are the same.

Question 4. Construct a DFA for all binary strings having 001 as a substring

Given: $\Sigma = \{0, 1\}$.

A string is accepted if it contains 001 anywhere in the string. Once 001 has appeared, any additional binary digits are allowed.

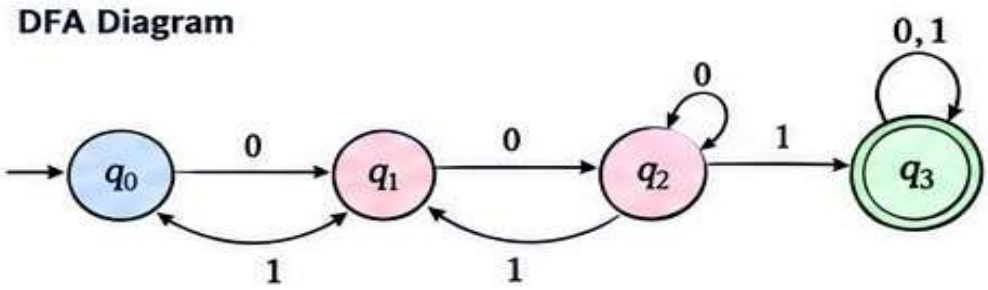
Examples: 001, 0010, 1001, 110010.

Step 1: Define the states

- q_0: No useful prefix of 001 has been matched.
- q_1: The first 0 has been matched.
- q_2: The substring 00 has been matched.
- q_3: The substring 001 has been found; this is the final state.

Step 2: Construct the transition table

State	Input 0	Input 1
$\rightarrow q_0$	q_1	q_0
q_1	q_2	q_0
q_2	q_2	q_3
$* q_3$	q_3	q_3



Step 4: Verify with an example

Take the string 1001.

$$q_0 \xrightarrow{1} q_0 \xrightarrow{0} q_1 \xrightarrow{0} q_2 \xrightarrow{1} q_3$$

Since $q_3$is a final state, 1001 is accepted.

Result: The DFA accepts all binary strings containing 001 as a substring.

Note: The uploaded assessment does not show a separate NFA diagram for Question 4, so this is the standard DFA constructed directly for the stated language.

Question 5. Find ε-closure of each state and construct the equivalent DFA

Given: The NFA contains ε-transitions, where ε means that the automaton can move to another state without reading any input symbol.

From the given diagram:

- $q_0$has a self-loop on input 1.
- $q_1$has a self-loop on input 0.
- $q_2$has self-loops on 0 and 1.
- There is an ε-transition from $q_0$to q_1.
- There is an ε-transition from $q_1$to q_2.
- Initial state: q_0

- Final state: q_2

Step 1: Find the ε-closure of each state

The ε-closure of a state includes the state itself and every state reachable from it using zero or more ε-transitions.

1. ε-closure of q_0:

From q_0, we can reach $q_1$and then $q_2$without reading input.

$$\varepsilon\text{-closure}(q_0) = \{q_0, q_1, q_2\}$$

2. ε-closure of q_1:

From q_1, we can reach q_2.

$$\varepsilon\text{-closure}(q_1) = \{q_1, q_2\}$$

3. ε-closure of q_2:

There are no outgoing ε-transitions from q_2.

$$\varepsilon\text{-closure}(q_2) = \{q_2\}$$

Step 2: Construct the equivalent DFA

Define the DFA states:

- $A = \{q_0, q_1, q_2\}$
- $B = \{q_1, q_2\}$
- $C = \{q_2\}$

The initial state is A, because it is the ε-closure of q_0.

Calculate the transitions:

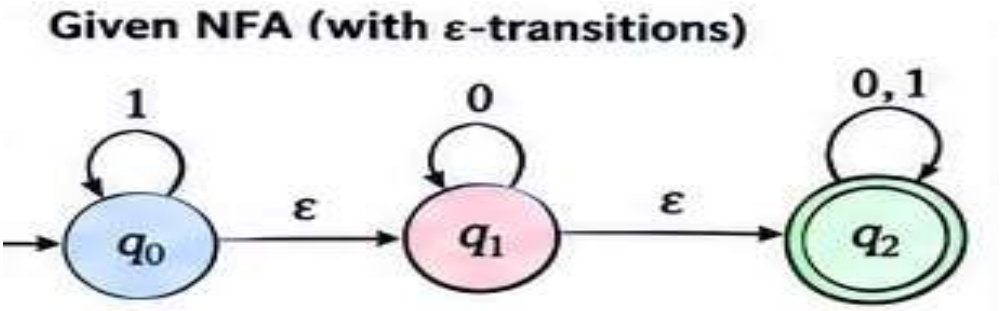
- From A, input 0 reaches B, and input 1 reaches A.

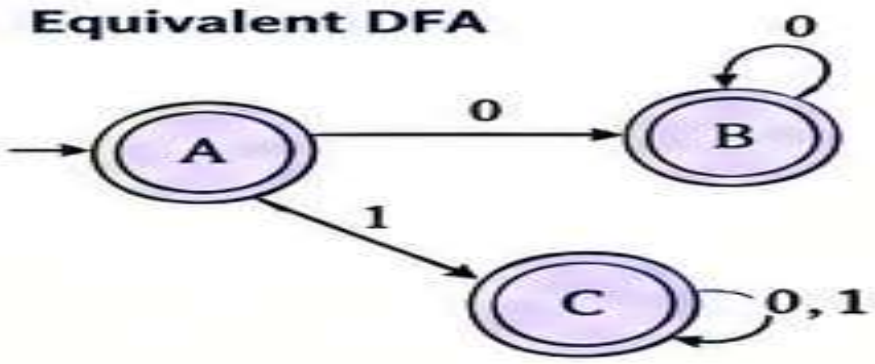
- From B, input 0 reaches B, and input 1 reaches C.
- From C, both 0 and 1 reach C

Step 3: DFA transition table

DFA state	0	1
$\rightarrow * A$	B	A
$* B$	B	C
$* C$	C	C

All three states are final because every state-set contains q_2, the final state of the original NFA.





Result

The ε-closures are:

$$\varepsilon\text{-closure}(q_0) = \{q_0, q_1, q_2\}$$
$$\varepsilon\text{-closure}(q_1) = \{q_1, q_2\}$$
$$\varepsilon\text{-closure}(q_2) = \{q_2\}$$

The equivalent DFA has three states, A, B, C, and accepts every binary string, including the empty string, because the original NFA can reach its final state $q_2$through ε-transitions before reading any input.